AMAZON KINDLE STORE REVIEW SETIMENT ANALYSIS

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Abstract

Today's world is getting increasingly computerised. In this digitalized world, e-commerce is gaining ground by bringing things to customers' doorsteps without requiring them to leave their homes. People nowadays rely on internet items, thus the value of a review and rating is increasing. A customer must read hundreds of reviews and ratings to understand a product before purchasing it. However, in this day and age of deep learning, sorting through hundreds of reviews and grading them would be lot easier if a model was used to polarise those reviews and learn from them. I ran sentiment analysis on a big dataset of Amazon kindle shop reviews to determine the good and poor ratings of things, and I noticed that positive ratings exceed negative ones.

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CHAPTER ONE

INTRODUCTION

1.1 Introduction

E-commerce demand is growing by the day. Because customers buy things without seeing or feeling them, customer voice ideas such as reviews and customer experiences are becoming increasingly significant in online retailers. If the organisation fails to address this client demand, it loses money due to a lack of strategic judgments.

This documenter chooses the dataset "Amazon Kindle Shop Reviews" and want to do an emotional analysis of the Amazon Kindle store reviews using Python. I will detect Good and Bad ratings using four different models, including Logistic Regression, SVC, and Random Forest, as well as three distinct algorithms, including Accuracy, AUC score, and Precision.

Individuals, corporations, institutions, and governments are increasingly relying on social media data to guide their decisions. When a person wants to buy a product, he no longer just questions others around him, but also searches the Internet for comments, conversations, and other information about the product. An company can also acquire an appraisal of its products and services over the Internet. Similarly, governments may easily obtain public comment on their policies as well as learn about key occurrences in other nations. However, due to the growth of different websites, online finding and monitoring of opinion websites, as well as information extraction from them, remains a tough process. The ordinary human reader will struggle to discover relevant sites and extract and summarise remarks from them. As a result, an automated emotional analysis system is required.

1.2 Aim

To readily identify evaluations as good or bad in order to assist companies boost customer contentment and value their remarks in order to save the organisation money.

CHAPTER TWO

METHODOLOGY

2.1 Data Collection

I focused on sentiment analysis of Kindle Store reviews on Amazon for this project. This dataset was chosen since it is easier to buy and read a book with Kindle. Going to the bookstore and finding a book you like takes longer than reaching every book on your iPad.

The data is obtained from github. Dataset contains 982619 reviews from 2000 till 2014. The following are the dataset features: asin – ID of the product, helpful – helpfulness rating of the review, overall – rating of the product, reviewText – text of the review (heading), reviewTime – time of the review (raw), reviewerID – ID of the reviewer, reviewerName – name of the reviewer, summary – summary of the review (description), unixReviewTime – unix timestamp. This dataset provides text reviews on books written by amazon kindle users along with an explicit rating between 1-5.

2.2 Method of Analysis

The purpose of this notebook is to practise text pre-processing and constructing models to predict user sentiment about a product. Given the specified scale, the sentiment is separated by ratings, with ratings 3-5 indicating that the book was enjoyed (binary one), and ratings 1-2 indicating that it was hated (binary zero).

2.3 Understanding, Cleaning and Exploring Data

The first problem with this data is to clear it of extraneous components for modelling purposes; the following steps are taken to accomplish this aim

i I made a dataframe that simply holds the reviewText and the item's total score.

ii Punctuation was deleted since punctuation is useless when developing a text categorization model.

iii Non-alphabetic words were eliminated.

iv Because there are strings with several characters in the uppercase format, I lowercased all words. The characters do not appear in a better way in the output string. In that scenario, you must alter the string from capital to lowercase.

V I deleted stopwords to lower the dataset size, which reduces training time owing to the fewer tokens involved in the training.

Vi I lemmatized in order to reduce words to their simplest form.

Vii divided the ratings into distinct sentiments and concatenated the two new dataframes provide one dataframe with preprocessed text and labels.

2.4 Preparing Data to Modelling

As a benchmark for what performance may be attained, a brief train using the Multinomial Naïve Bayes method was utilised. There are three things I did before training the model.

i I balanced the information. Because there is a mismatch in ratings, the sentiment target labels are also imbalanced, and training a model with data that contains much more of one class drives the model to categorise all fresh samples to the majority class. There are a few options, but I choose to downsample the majority class to the minority class because there are so many cases in this dataset.

ii I created the vocabulary. This is accomplished by utilising the CountVectorizer or TfidfVectorizer to convert the words in each review to a vector with elements that are either a word frequency or a weight. This approach, known as the bag of words method, does not take word ordering into account.

iii Tuned the vectorizer's min_df parameter and the alpha in the multinomial Naïve Bayes, ran param grid, and iterated throughout the param grid.

2.5 Modeling

To begin, many measures were utilised, including Accuracy, AUC score, and precision. Following that, various models such as Logistic Regression, SVC, and Random Forest were trained to determine whether they would outperform the TfidfVectorizer. Moreso I used Neural network which is LSTM

CHAPTER THREE

ANALYSIS AND RESULTS

3.1 Analysing Dataset information

The dataset consist of 10 columns, 982619 asin (ID of the product), 982619 helpful reviews, 982619 overall reviews, 982619 review texts, 982597 review time, 982619 reviewer ID, 978803 reviewer name, 982618 summary of review and 982619 unix review time. **Figure 3.1** shows the data information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 982619 entries, 0 to 982618
Data columns (total 10 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   Unnamed: 0      982619 non-null  int64
 1   asin            982619 non-null  object
 2   helpful         982619 non-null  object
 3   overall         982619 non-null  int64
 4   reviewText      982597 non-null  object
 5   reviewTime      982619 non-null  object
 6   reviewerID      982619 non-null  object
 7   reviewerName    978803 non-null  object
 8   summary         982618 non-null  object
 9   unixReviewTime  982619 non-null  int64
dtypes: int64(3), object(7)
memory usage: 75.0+ MB
```

Figure 3.1: Data Information

I checked for missing values when analysing my dataset, and 22 review text and 3816 reviewer ID were missing. The output of missing values in my dataset is shown in **Figure 3.2** below.

```
Unnamed: 0          0
asin                0
helpful             0
overall             0
reviewText         22
reviewTime          0
reviewerID          0
reviewerName     3816
summary             1
unixReviewTime      0
dtype: int64
```

Figure 3.2: Missing values

I plotted the amount of reviews against the year to see how many there were each year, and it looks that almost all of the reviews were from 2013 and 2014. **Figure 3.3** illustrates a depiction of the number of reviews received each year.

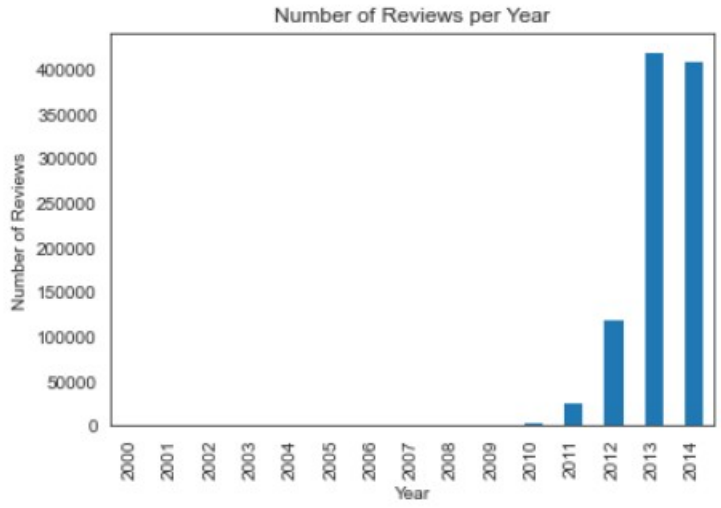


Figure 3.3: Number of Reviews per year

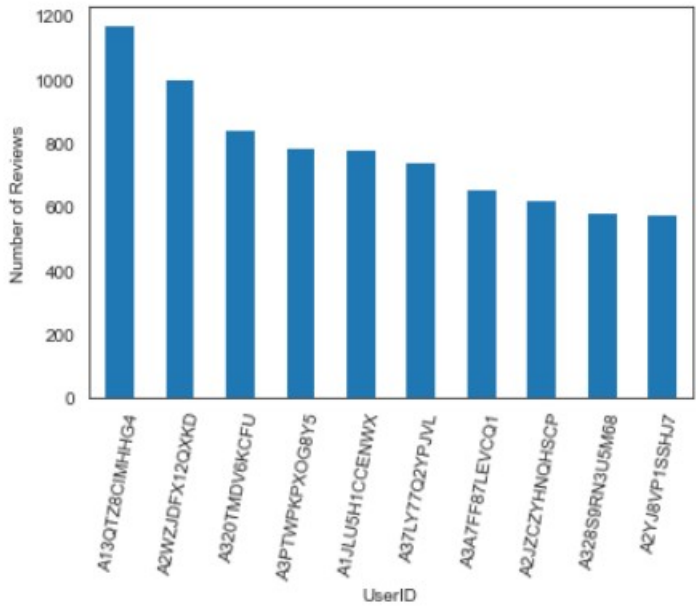


Figure 3.4: Top 10 users base on the number of reviews made

I investigated how many ratings I have per scale. There are more positive ratings than negative ratings. Figure 3.5 depicts a plot of the number of Good Ratings versus Bad Ratings.

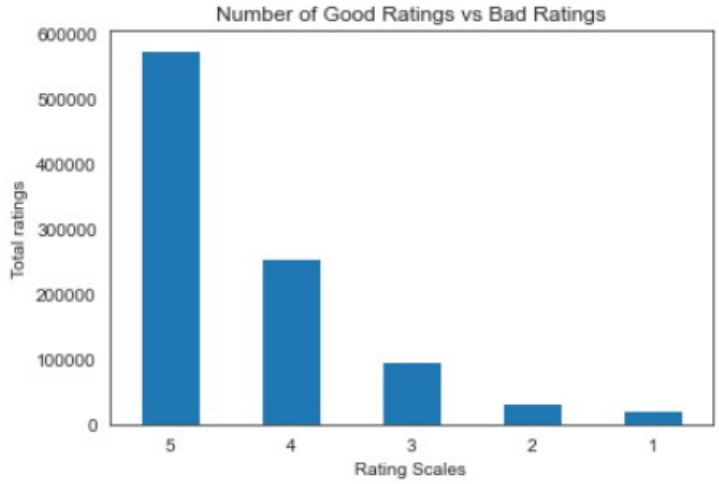


Figure 3.5: Number of Good Ratings vs Bad Ratings

3.2 Modelling

As a consequence, training on a Multinomial Naïve Bayes appears to yield an accuracy of 91% without tuning any parameters, regardless of whether the vectorizer is CountVectorizer or TfidfVectorizer, as shown in **Figure 3.7**. There are more features than samples in the training examples, based on their size. The min_df option can be used to limit the amount of features.

However, the vocabulary contains around 80000 terms based on a sample of 40000 observations, as illustrated in **Figure 3.8**. Finding the proper value for min df is required to decrease the amount of word features. If an integer is used for this option, CountVectorizer and TfidfVectorizer need a word to appear in a minimum number of documents in the total corpus. If a float is passed, the word must fall within this percentage of the corpus. When instantiating the Naïve Bayes algorithm, another parameter that may be tweaked is the alpha term. This option prevents the likelihood probability from being set to 0 simply because a word did not appear in the training instances. These settings will be tuned to determine whether the Naïve Bayes method can be improved further.

Finally, after tuning the parameters using K-Fold cross validation, the model still delivered a testing accuracy of 87% but with around 3600 words compared to the prior 80000 with a min df of.001, as seen in **Figure 3.9**. Although.00001 was determined to be the optimum min_df, this value only increased training accuracy, therefore it was tuned down to further minimise the amount of word features while retaining testing accuracy. This model was trained using a TfidfVectorizer, as will the rest of the notebook.

Figure 3.6 displays the number of good and bad ratings. A good rating is represented by 1 (925449), while a bad rating is represented by 0 (57148). This data confirms my conclusion in **Figure 3.5** that Good ratings outnumber Bad ratings.

```
rev.overall.value_counts()
```

```
1    925449
0     57148
Name: overall, dtype: int64
```

Figure 3.6: The number of good and bad rating

```
#testing the model with CountVectorizer
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = .3, random_state = 42, stratify = y)

naive = MultinomialNB()
naive.fit(X_train, y_train)
print('Training Accuracy w/ CountVectorizer: {:.2f}'.format(naive.score(X_train, y_train)))
print('Testing Accuracy w/ CountVectorizer: {:.2f}'.format(naive.score(X_test, y_test)))
```

```
Training Accuracy w/ CountVectorizer: 0.91
Testing Accuracy w/ CountVectorizer: 0.87
```

Figure 3.7: Result using Count Vectorizer

```
X.shape
```

```
(40000, 84510)
```

Figure 3.8: Vocabulary and sampling observations.

```
X_train.shape
```

```
(28000, 3690)
```

Figure 3.9: After tuning the parameters using K-Fold cross validation

ROC curves are commonly used in binary classification to investigate a classifier's output. To apply the ROC curve and ROC area to multi-label classification, the result must be binarized. One ROC curve can be generated for each label, but a ROC curve can also be drawn by treating each member of the label indication matrix as a binary prediction (micro-averaging).

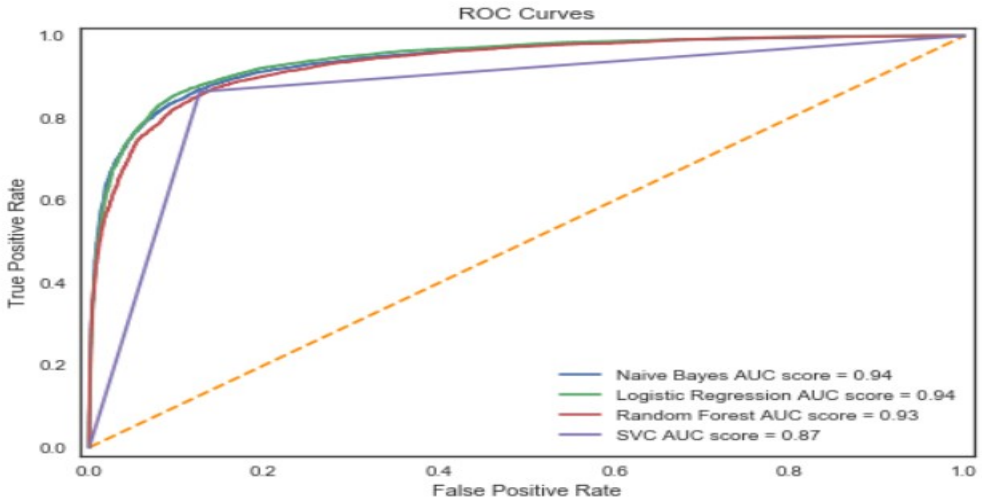


Figure 3.10: ROC Curves

3.3 Neural Network

LSTM

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 embedding (Embedding)       (None, 200, 32)           27066272

 spatial_dropout1d (SpatialD  (None, 200, 32)          0
 ropout1D)

 lstm (LSTM)                 (None, 50)                16600

 dropout (Dropout)           (None, 50)                0

 dense (Dense)               (None, 1)                 51

=================================================================
Total params: 27,082,923
Trainable params: 27,082,923
Non-trainable params: 0
_________________________________________________________________
None
```

```
history = model.fit(padded_sequence,sentiment_label[0],validation_split=0.25,epochs=3,batch_size=64)
```

```
Epoch 1/3
11515/11515 [==============================] - 7600s 658ms/step - loss: 9.8582e-04 - accuracy: 0.9999 - val_loss: 4.0198 - v
al_accuracy: 0.7674
Epoch 2/3
11515/11515 [==============================] - 17904s 2s/step - loss: 1.6266e-08 - accuracy: 1.0000 - val_loss: 5.2556 - val
_accuracy: 0.7674
Epoch 3/3
11515/11515 [==============================] - 87565s 8s/step - loss: 3.8107e-10 - accuracy: 1.0000 - val_loss: 5.6534 - val
_accuracy: 0.7674
```

CHAPTER FOUR

CONCLUSION AND FUTURE WORK

4.1 Conclusion

Overall, when numerous measures like as accuracy, AUC score and precision were used, these three algorithms produced similar results to Naïve Bayes.

It is difficult to make predictions using simply raw text as input. As a result, we discovered that the pre-processing stage was critical to the overall NLP process. To achieve better results, accuracy, and have the machine accept all text as tokens, data must be pre-processed thoroughly, taking into account the sort of contents present. The most critical aspect is to be able to extract the appropriate characteristics from the provided data source. This type of data is frequently used as a supplement to extract more learning features and improve the prediction ability of models. And the user may anticipate if the supplied rating is excellent or negative.

4.2 Future works

In the future, the work might be expanded to do multi-class categorization of reviews, which would offer the customer with a more defined type of review and therefore better judgement of the product. It may also be used to anticipate the positive and negative terms in a product review. This will give users with a trustworthy review because occasionally the product's review and the mood of the rating do not correspond. The suggested work extension would be highly advantageous to the e-commerce business since it will increase consumer happiness and confidence.

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